Echelon form and Rank of a Matrix:

Q.1 Define Rank and Echelon form of a matrix. Reduce the following matrices to its Echelon form and hence find its rank:

$$(a)\ \begin{bmatrix} 1 & 1 & -1 & 1 \\ 1 & -1 & 2 & -1 \\ 3 & 1 & 0 & 1 \end{bmatrix} \qquad (b)\ \begin{bmatrix} 0 & 1 & 2 & -2 \\ 4 & 0 & 2 & 6 \\ 2 & 1 & 3 & 1 \end{bmatrix} \qquad (c)\ \begin{bmatrix} 3 & 2 & 5 & 7 & 12 \\ 1 & 1 & 2 & 3 & 5 \\ 3 & 3 & 6 & 9 & 15 \end{bmatrix}$$

$$(d)\ \begin{bmatrix} 2 & 3 & -1 & -1 \\ 1 & -1 & -2 & -4 \\ 3 & 1 & 3 & -2 \\ 6 & 3 & 0 & -7 \end{bmatrix} \qquad (e)\ \begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix} \qquad (f)\ \begin{bmatrix} 3 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 4 \\ 2 & 3 & 3 & 4 & 6 \\ 1 & 2 & 4 & 1 & 8 \end{bmatrix}$$

$$(g)\ \begin{bmatrix} 1 & 2 & 1 & 0 \\ 3 & 2 & 1 & 2 \\ 2 & -1 & 2 & 5 \\ 5 & 6 & 3 & 2 \\ 1 & 3 & -1 & -3 \end{bmatrix} \qquad (h)\ \begin{bmatrix} 3 & -2 & 0 & -1 & -7 \\ 0 & 2 & 2 & 1 & -5 \\ 1 & -2 & -3 & -2 & 1 \\ 0 & 1 & 2 & 1 & -6 \end{bmatrix} \qquad (i)\ \begin{bmatrix} 3 & -6 & 4 & -3 & 2 \\ 2 & -4 & 3 & 1 & 0 \\ 0 & 1 & -1 & 3 & 1 \\ 4 & -7 & 4 & -4 & 5 \end{bmatrix}$$

$$(j)\ \begin{bmatrix} 2 & 3 & 4 & 5 \\ 3 & 4 & 5 & 6 \\ 4 & 5 & 6 & 7 \\ 9 & 10 & 11 & 12 \end{bmatrix} \qquad (k)\ \begin{bmatrix} 2 & -1 & 3 & 1 \\ 1 & -4 & -2 & 1 \\ 5 & 2 & 4 & 3 \end{bmatrix}$$

System of Non-homogeneous Linear Equations:

Q.2 Test for consistency and if consistent, solve the following system of equations:

$$(a)\ \begin{aligned} 2x_1 + x_2 + 2x_3 + x_4 &= 6 \\ 6x_1 - 6x_2 + 6x_3 + 12x_4 &= 36 \\ 4x_1 + 3x_2 + 3x_3 - 3x_4 &= -1 \\ 2x_1 + 2x_2 - x_3 + x_4 &= 10 \end{aligned} \qquad (b)\ \begin{aligned} x + y + z &= 3 \\ 2x - y + 3z &= 1 \\ 4x + y + 5z &= 2 \\ 3x - 2y + z &= 4 \end{aligned} \qquad (c)\ \begin{aligned} 2x - y - z &= 2 \\ x + 2y + z &= 2 \\ 4x - 7y - 5z &= 2 \end{aligned}$$

Q.3 (a) For what values of k the equations: $x+y+z=1;\ 2x+y+4z=k;$ $4x+y+10z=k^2$ have infinite number of solutions? Hence find solutions.

(b) Show that the system $3x+4y+5z=\alpha;\quad 4x+5y+6z=\beta;\ 5x+6y+7z=\gamma$ is consistent only when α,β,γ are in arithmetic progression.

(c) For what values of k the following set of equations possesses infinitely number of solution and hence find solution for respective value of k, $2x-3y+6z-5t=3;$ $y-4z+t=1;\ 4x-5y+8z-9t=k.$

(d) For what values of k, the set of equations $2x-3y+6z=3;\quad y-4z=1;$ $4x-5y+8z=k$ have infinite number of solutions. Hence find solutions.

(e) Investigate the values of λ and μ so that the equations: $2x+3y+5z=9;$

$7x+3y-2z=8;\ 2x+3y+\lambda z=\mu$ have

(a) no solution (b) a unique solution

(c) an infinite number of solutions.

(f) Investigate for what values of λ and μ the system of simultaneous equations,

$x+y+z=6;\ x+2y+3z=10;\ x+2y+\lambda z=\mu$ have

(a) no solution (b) an infinite number of solutions.

(g) Determine the value of λ for which the equations, $3x_1+2x_2+4x_3=3;$ $5x_1+4x_2+6x_3=15$, $x_1+x_2+x_3=\lambda;$ are consistent. Find also the corresponding solution.

(h) Show that the system of equations, $3x+4y+5z=a;\quad 4x+5y+6z=b;$ $5x+6y+7z=c$ will be consistent only if $a+c=2b$.

(i) Determine the value of λ for which the equations, $x+2y+z=3;\ x+y+z=\lambda$, $3x+y+3z=\lambda^2;$ are consistent and solve them for the values of λ.

Linearly Dependence and Linearly Independence:

Q.4 Test the following vectors for linearly dependence and find a relation between them if dependent:

(a) $(1,\ 2,\ 4),\ (2,\ -1,\ 3),\ (0,\ 1,\ 2),\ (-3,\ 7,\ 2).$

(b) $(1,\ 2,\ -1,\ 0),\ (1,\ 3,\ 1,\ 2),\ (4,\ 2,\ 1,\ 0),\ (6,1,\ 0,\ 1).$

(c) $(1,\ -1,\ 1),\ (2,\ 1,\ 1),\ (3,\ 0,\ 2).$

(d) $(3,\ 7,\ -2,\ 4),\ (2,\ 3,\ -1,\ 2),\ (10,\ 20,\ -6,\ 12).$

(e) $(2,\ -1,\ 3,\ 2),\ (3,\ -5,\ 2,\ 2),\ (1,3,\ 4,\ 2).$

(f) $(3,\ 1,\ -4),\ (2,\ 2,\ -3),\ (0,\ -4,\ 1).$

(g) $(2,\ 3,\ 4,\ -2),\ (-1,\ -2,\ -2,\ 1),\ (1,1,\ 2,\ -1).$

(h) $(2,\ -1,\ 3,\ 2),\ (1,\ 3,\ 4,\ 2),\ (3,\ -5,\ 2,\ 2).$

(i) $(1,\ 3,\ 4,\ 2),\ (3,\ -5,\ 2,\ 2),\ (2,\ -1,\ 3,\ 2).$

Eigen Values and Eigen Vectors:

Q.5 Find the Eigen values and Eigen vectors of the following matrices:

(i) $\begin{bmatrix} 14 & -10 \\ 5 & -1 \end{bmatrix}$ (ii) $\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ (iii) $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

(iv) $\begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$ (v) $\begin{bmatrix} 4 & 2 & -2 \\ -5 & 3 & 2 \\ -2 & 4 & 1 \end{bmatrix}$ (vi) $\begin{bmatrix} 3 & 1 & 1 \\ 1 & 3 & -1 \\ 1 & -1 & 3 \end{bmatrix}$

(vii) $\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$ (viii) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{bmatrix}$ (ix) $\begin{bmatrix} 4 & -1 & 1 \\ -1 & 4 & -1 \\ 1 & -1 & 4 \end{bmatrix}$

(x) $\begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$ (xi) $\begin{bmatrix} 3 & -1 & 1 \\ -1 & 3 & -1 \\ 1 & -1 & 3 \end{bmatrix}$ (xii) $\begin{bmatrix} 1 & -6 & -4 \\ 0 & 4 & 2 \\ 0 & -6 & -3 \end{bmatrix}$

$(xiii)$ $\begin{bmatrix} -9 & 4 & 4 \\ -8 & 3 & 4 \\ -16 & 8 & 7 \end{bmatrix}$ (xiv) $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$ (xv) $\begin{bmatrix} -17 & 18 & -6 \\ -18 & 19 & -6 \\ -9 & 9 & -2 \end{bmatrix}$

(xvi) $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$ $(xvii)$ $\begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$ $(xviii)$ $\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

(xix) $\begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 1 \\ -2 & 1 & -1 \end{bmatrix}$